

Homework 6

Oct 22, 2025

1 (30'). Define a *two-dimensional* Turing machine to be a TM where each of its tapes is an infinite grid (and the machine can move not only Left and Right but also Up and Down).

Show that if language L can be decided in time $T(n)$ by a two-dimensional TM then $L \in \text{DTIME}(T(n)^2)$.

2 (40'). Define a *one-tape two-head Turing machine* as a Turing machine equipped with a single tape but two independent heads operating on it. Each head can read, write, and move simultaneously and independently of the other. Both heads may read the same tape cell at the same time. Conflicts are resolved as follows:

- If both heads attempt to write to the same cell simultaneously, the first head goes first, and the second head overwrites any previous content.
- If the first head tries to write while the second head tries to read the same cell, the first head goes first, and the second head reads the updated content.
- If the first head tries to read while the second head tries to write the same cell, the first head goes first, reading the old content, and then the second head updates the content.

Show that if language L can be decided in time $T(n)$ by a one-tape two-head Turing machine, then $L \in \text{DTIME}(T(n)^2)$.

3. (30') Prove the number of multitape Turing machines is countable.